

CAN Bus Based Vehicle Anti-Theft System with Engine Lockdown and GPS Alert

*A Project Report Submitted in Partial
Fulfillment of the Requirements for the
Completion of the*

ePBL INTERNSHIP

IN

EMBEDDED SYSTEMS

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Abstract

Automotive electronics and in-vehicle communication systems have increased vehicle capabilities and improved safety and convenience. However, these technologies have created new safety risks for vehicles – specifically, security threats from cyberattacks are abundant on the Controller Area Network (CAN) bus that connects Electronic Control Units (ECUs). Since the CAN bus lacks security capabilities for verifying or encrypting messages, it is vulnerable to attacks and unauthorised access.

The goal of this project is to design a Vehicle Anti-Theft System based on the CAN bus with Engine Lockdown (EL) and GPS Alert. This system will work together with other technologies to provide an overall vehicle security solution through intrusion detection, real-time communication, location tracking, and control. An infrared (IR) sensor will be used to detect when someone enters an area where there should not be any non-automotive traffic; this information will be sent through the CAN bus as well as a LoRa (Long Range) communication module from each ESP32 Node.

In the event of unexpected activity on a vehicle, an immediate alert is sent to the owner via a GSM module as soon as the activities occurred. Concurrently, a GPS module verifies the current location of the vehicle in order to provide real time tracking. On the occasion that a situation is determined as being critical, a relay module is utilized to perform an engine lock down in order to prevent the use of the vehicle without the owner's consent. These features combine to provide an effective solution for both detecting and preventing vehicle theft.

Furthermore, from the experimental data obtained, it is evident from the research that the proposed method possesses the ability to detect an unauthorized entry attempt, generate alerts in a timely manner, and effectively restrict the use of the vehicle if required. This system offers a reliable, cost-effective, and scalable solution to improve vehicle security. The combined effort of both CAN bus monitoring and tracking capabilities offered by the proposed method addresses the limitations of current systems and offers additional means for creating sophisticated automotive security measures.

Keywords: CAN Bus, Vehicle Security, GPS Tracking, GSM Alerts, Engine Lockdown, Intrusion Detection

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Abbreviations

Abbreviation	Description
CAN	Controller Area Network
ECU	Electronic Control Unit
GPS	Global Positioning System
IoT	Internet of Things
ESP32	Espressif 32-bit Microcontroller
LoRa	Long Range Communication
IDS	Intrusion Detection System
IR	Infrared
DoS	Denial of Service
MCU	Microcontroller Unit
CANH	CAN High Line
CANL	CAN Low Line
TX	Transmit
RX	Receive
GPIO	General Purpose Input Output
UART	Universal Asynchronous Receiver Transmitter
SPI	Serial Peripheral Interface
IP	Internet Protocol
IDE	Integrated Development Environment
PCB	Printed Circuit Board
CAV	Connected and Autonomous Vehicle
ABS	Anti-lock Braking System
ADAS	Advanced Driver Assistance Systems
SCC	Smart Cruise Control
VCE	Vardhaman College of Engineering

CHAPTER 1

Introduction

1.1 Introduction

Modern vehicles are increasingly connected to advanced electronic systems and communication networks. While these technologies improve comfort, safety, and automation, they also expose vehicles to potential cyber-security threats. One of the most critical communication systems inside a vehicle is the Controller Area Network (CAN) bus, which acts as the backbone for communication between different electronic control units (ECUs). These ECUs manage essential functions such as engine control, braking systems, steering, and other critical safety operations.

However, the CAN bus was originally designed for reliability and efficiency, rather than security. As a result, it lacks built-in authentication and encryption mechanisms, making it vulnerable to malicious attacks. Cyber-attackers can exploit these vulnerabilities to inject false messages, manipulate vehicle behavior, or gain unauthorized control over vehicle functions. Traditional security mechanisms often rely on basic monitoring techniques, which are not always capable of detecting sophisticated or real-time intrusions within the CAN network.

To address these challenges, integrating a CAN bus intrusion detection system with a GPS-based lockdown mechanism can significantly improve vehicle security. The intrusion detection system continuously monitors CAN bus traffic and identifies abnormal communication patterns that may indicate a cyber-attack or unauthorized access. When suspicious activity is detected, the system can trigger a security response, such as alerting the vehicle owner and activating a lockdown feature to restrict certain vehicle operations.

In addition to intrusion detection, the integration of GPS tracking allows the vehicle's location to be monitored in real time. This feature is particularly

useful in the case of theft or unauthorized vehicle use. The system can immediately notify the owner through alerts and provide precise location information, enabling quick response and recovery.

Overall, this integrated approach enhances vehicle safety by combining real-time attack detection, location tracking, and automated security responses. By preventing unauthorized access, tracking vehicle movements, and instantly notifying the owner, the proposed system offers a more robust solution to protect modern connected vehicles from emerging cyber threats.

1.2 Background and Motivation

In recent years, the rapid advancement of automotive electronics and communication technologies has significantly transformed modern vehicles. Contemporary vehicles are equipped with multiple Electronic Control Units (ECUs) that communicate through the Controller Area Network (CAN) bus, enabling efficient control of critical functions such as engine management, braking, and steering. While this interconnected architecture improves performance, safety, and user convenience, it also introduces serious security vulnerabilities.

The internal structure of a modern vehicle, consisting of multiple ECUs connected through different CAN subnetworks, is illustrated in Figure 1.1. This complex communication framework enables seamless interaction between various subsystems, but at the same time increases the potential attack surface within the vehicle network.

The CAN bus protocol, originally designed for reliability and real-time communication, lacks essential security features such as authentication and encryption. As a result, it is highly susceptible to cyber-attacks, where unauthorized users can inject malicious messages, manipulate vehicle behavior, or gain control over critical systems. Since all ECUs are interconnected, a vulnerability in one part of the network can potentially affect the entire vehicle system.

In addition to cybersecurity concerns, the increasing rate of vehicle theft and unauthorized access highlights the need for more advanced and integrated security solutions. Traditional vehicle anti-theft mechanisms such as alarms

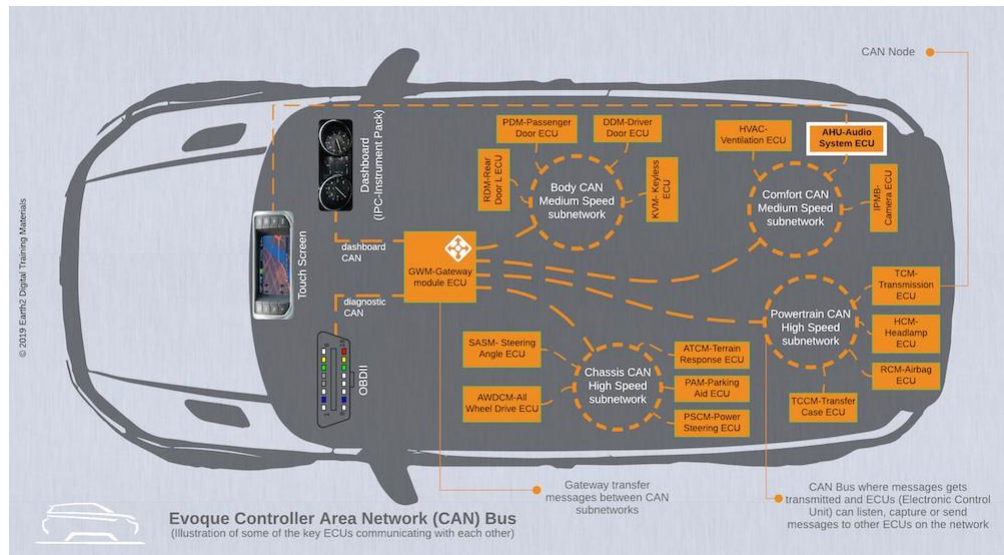


Figure 1.1: Illustration of ECU Interconnections and CAN-Based Communication in Modern Vehicles

and immobilizers are no longer sufficient to counter modern threats, especially those involving electronic and network-based attacks. Existing systems often focus on individual functionalities such as GPS tracking or GSM-based alerting, but fail to provide a comprehensive solution that combines real-time detection, communication, tracking, and control.

Motivated by these challenges, this project aims to develop an integrated vehicle anti-theft system that combines CAN bus monitoring, intrusion detection, GPS-based tracking, GSM alert notifications, and engine lockdown mechanisms. By continuously monitoring vehicle activity, detecting suspicious behavior, and enabling immediate response through alerts and remote immobilization, the proposed system enhances both vehicle safety and security.

The motivation behind this work is to bridge the gap between traditional security approaches and modern technological requirements by designing a cost-effective, reliable, and scalable solution. This integrated approach not only helps in preventing vehicle theft but also ensures quick recovery and improved protection against emerging cyber-physical threats in connected vehicles.

1.3 Objectives of the Project Work

The objectives guide the development of the system and ensure that the project focuses on solving the problem of vehicle theft through an efficient

and reliable technological solution.

- To develop a CAN Bus based vehicle security system that enables communication between the vehicle's electronic components and the anti-theft module.
- To detect unauthorized access or suspicious activity using sensors and CAN network monitoring.
- To implement GPS-based real-time vehicle tracking for location monitoring.
- To develop a GSM alert system that sends instant notifications to the vehicle owner during theft attempts.
- To integrate an engine lockdown mechanism to remotely disable the vehicle and prevent unauthorized use.

1.4 Organization of the Report

The final section of Chapter 1 provides an outline of the structure of the entire report. It explains the organization and flow of the chapters and gives the reader an understanding of how the content will be presented. This section can be a simple list or paragraph summarizing each chapter's main points.

- Provide a brief description of what each chapter will cover.
- Typically, the structure would include:

- **Chapter 2:** Literature Review

This chapter reviews existing research and technologies related to vehicle security systems. It discusses various approaches such as CAN Bus intrusion detection, GPS-based tracking, GSM alert systems, and engine immobilization techniques. The chapter also identifies limitations in existing systems and highlights the research gaps addressed by the proposed work.

- **Chapter 3:** System Design and Architecture

This chapter presents the overall design and architecture of the proposed vehicle anti-theft system. It explains the system structure

using block and circuit diagrams, and describes how different hardware components such as ESP32 microcontrollers, CAN transceivers, sensors, and control modules are interconnected. The chapter also discusses the software aspects involved in system development.

– **Chapter 4:** Implementation and Methodology

This chapter focuses on the implementation and working of the proposed vehicle anti-theft system. It explains the methodology adopted for system operation, including intrusion detection, CAN bus communication, and control mechanisms. The chapter also presents the flowchart and describes the system operation in detail, highlighting how different modules interact in real time.

– **Chapter 5:** Results and Discussion

This chapter presents the results obtained from the implementation and testing of the system. It analyzes the performance of each module, including intrusion detection, alert generation, tracking accuracy, and engine lockdown response. The chapter also evaluates the system based on factors such as reliability, efficiency, cost, and safety.

– **Chapter 6:** Conclusions and Recommendations

This chapter summarizes the overall outcomes of the project and highlights the effectiveness of the proposed system in enhancing vehicle security. It also suggests possible improvements and future enhancements, such as mobile application integration, cloud connectivity, and advanced intrusion detection techniques.

– **Chapter 7:** Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

This chapter explains how the project contributes to academic program outcomes and aligns with sustainable development goals. It highlights the technical skills, problem-solving abilities, and societal impact achieved through the project.

CHAPTER 2

Literature Survey

2.1 Introduction

In recent years, the automotive industry has undergone a significant transformation with the integration of advanced electronic systems and communication technologies [1]. Modern vehicles are no longer purely mechanical machines; instead, they are complex embedded systems consisting of numerous Electronic Control Units (ECUs) that work collaboratively to ensure efficient operation, safety, and user comfort. These ECUs communicate with each other through in-vehicle networks, among which the Controller Area Network (CAN) bus is the most widely adopted protocol due to its reliability, low cost, and real-time communication capabilities [2].

The CAN bus enables seamless data exchange between various subsystems such as engine control, braking systems, transmission units, airbag systems, and infotainment modules. This interconnected architecture has significantly improved vehicle performance and automation. However, the increasing dependence on network-based communication has also introduced serious security challenges. Since the CAN protocol was originally designed without considering cybersecurity aspects, it lacks essential features such as authentication, encryption, and access control, making it vulnerable to attacks [3].

With the rapid growth of connected vehicles and external interfaces such as Bluetooth, Wi-Fi, GPS, and mobile networks, the attack surface of vehicles has expanded considerably. Attackers can exploit these entry points to gain unauthorized access to the vehicle's internal network and manipulate critical functions. Potential attacks include message spoofing, replay attacks, and Denial-of-Service (DoS) attacks, which can compromise vehicle safety [4].

To address these challenges, researchers have focused on developing Intrusion Detection Systems (IDS) specifically designed for in-vehicle networks.

The base paper, “In-Vehicle Network Intrusion Detection System Using CAN Frame-Aware Features,” proposes an advanced IDS that utilizes the inherent characteristics of CAN messages to detect anomalies [5]. By analyzing parameters such as message identifiers, timing intervals, and data patterns, the system can effectively distinguish between normal and malicious behavior.

Although such IDS-based approaches significantly enhance the security of CAN networks, they primarily focus on detection rather than prevention. In practical scenarios, an effective vehicle security system must not only detect intrusions but also provide immediate response mechanisms such as alert generation, vehicle tracking, and engine immobilization.

Therefore, this chapter explores existing research with a primary focus on CAN-based intrusion detection systems, along with supporting technologies such as GPS tracking and GSM communication [6, 7]. The insights gained from this study serve as the foundation for developing an integrated vehicle anti-theft system.

2.2 Review of Prior Research

The security of in-vehicle networks has emerged as a critical area of research due to the increasing integration of electronic control systems in modern vehicles. The Controller Area Network (CAN) bus, being the backbone of in-vehicle communication, has been widely studied for its vulnerabilities and security limitations [2, 8]. Although the CAN protocol offers efficient and reliable communication, its lack of built-in security mechanisms such as authentication and encryption makes it highly susceptible to various cyber-attacks [9].

Early research in this domain primarily focused on understanding the nature of CAN bus communication and identifying potential attack vectors. Researchers demonstrated that attackers could exploit the broadcast nature of CAN messages to inject malicious data into the network. Attacks such as message spoofing, replay attacks, and Denial-of-Service (DoS) attacks were identified as major threats capable of disrupting vehicle functionality and compromising safety [3, 10]. These findings highlighted the urgent need for

effective monitoring and detection mechanisms.

Initial solutions to these challenges involved rule-based intrusion detection systems. These systems relied on predefined rules and thresholds to detect abnormal behavior in CAN traffic. Parameters such as message frequency, identifier consistency, and timing intervals were analyzed to identify suspicious activities. While these approaches were effective in detecting known attack patterns, they lacked adaptability and were unable to handle dynamic or previously unseen attacks [11].

To address these limitations, researchers began exploring statistical and machine learning-based approaches. These methods utilized features extracted from CAN messages to train classification models capable of distinguishing between normal and malicious traffic [12, 4]. Machine learning techniques improved detection accuracy and provided greater flexibility compared to rule-based systems. However, they often required labeled datasets and struggled with capturing temporal dependencies in sequential data.

With advancements in artificial intelligence, deep learning-based intrusion detection systems were introduced. Recurrent neural networks such as Long Short-Term Memory (LSTM) models were widely used to analyze sequential CAN data and capture temporal patterns [12]. These models improved the detection of complex attack scenarios by learning patterns over time. However, LSTM-based systems sometimes faced challenges such as high computational complexity and dependency on large datasets.

To further enhance detection performance, hybrid deep learning models such as CNN-LSTM were proposed. These models combine convolutional neural networks for spatial feature extraction with LSTM networks for temporal analysis, providing a more comprehensive understanding of CAN traffic patterns [13, 14]. Such hybrid approaches demonstrated improved accuracy in detecting both known and unknown attacks, making them suitable for real-time applications.

In addition to supervised learning methods, unsupervised and anomaly detection techniques have also gained attention. Algorithms such as Isolation Forest have been used to detect anomalies in CAN traffic without requiring

labeled data [15]. These methods are particularly useful in real-world scenarios where obtaining labeled datasets is challenging. They can identify unusual patterns that deviate from normal behavior, enabling the detection of previously unseen attacks.

The base paper further advances this research by introducing CAN frame-aware feature extraction techniques [5]. Unlike traditional methods that rely solely on statistical features, this approach considers both structural and temporal characteristics of CAN messages. By analyzing message identifiers, payload content, and timing intervals, the system achieves higher accuracy and improved detection performance.

Apart from intrusion detection, researchers have also explored the integration of vehicle security systems with communication and monitoring technologies. IoT-based vehicle security solutions have been proposed to enable real-time monitoring and remote control of vehicles [16, 17]. These systems utilize wireless communication technologies to transmit data and provide user access through web or mobile interfaces.

Similarly, GPS-based tracking systems have been widely used to monitor vehicle location and assist in theft recovery [6, 18]. GSM-based alert systems have also been implemented to notify vehicle owners in case of unauthorized access or suspicious activity [7, 19]. These technologies enhance the overall effectiveness of vehicle security systems by providing real-time information and user interaction.

Recent studies have also explored the use of embedded platforms such as ESP32 for implementing low-cost and efficient vehicle security solutions [esp32'wireless2021, 20]. These platforms enable wireless communication, real-time processing, and integration with IoT systems, making them suitable for practical implementations.

Furthermore, long-range communication technologies such as LoRa have been investigated for vehicle monitoring applications, providing reliable data transmission over extended distances with low power consumption [21]. This enhances the scalability and applicability of vehicle security systems in real-world environments.

Despite significant advancements in intrusion detection and vehicle security

technologies, most existing solutions focus on individual functionalities. Many systems are limited to detection without providing mechanisms for response or prevention. For instance, while IDS can identify attacks, they do not typically include features such as engine immobilization or automated control actions [22].

Furthermore, integration between different components such as detection, communication, tracking, and control is often lacking. This results in fragmented solutions that are not fully effective in real-world scenarios. Practical vehicle security systems require a unified approach that combines multiple functionalities into a single framework [23].

Another limitation observed in existing research is the lack of scalability and cost-effectiveness. Some advanced detection systems require high computational resources, making them unsuitable for implementation in low-cost embedded systems [24]. Therefore, there is a need for solutions that are both efficient and feasible for real-world deployment.

Overall, the review of prior research indicates that while significant progress has been made in identifying and detecting cyber-attacks in CAN-based systems, there is still a gap in developing integrated and practical solutions. These observations emphasize the need for a comprehensive system that combines intrusion detection, real-time communication, tracking, and control mechanisms to enhance vehicle security effectively.

2.3 Identified Research Gaps

1. Lack of Preventive Mechanisms

The base paper primarily focuses on detecting intrusions within the CAN bus network. While detection is important, the system does not provide mechanisms to prevent or respond to attacks. This highlights the need for integrating response mechanisms such as control actions.

2. Absence of Real-Time Alert System

The intrusion detection system does not include a real-time alert mechanism. In practical systems, instant alerts using communication technologies such as GSM are essential for user awareness.

3. No Integration with Vehicle Tracking

The base paper does not include GPS-based tracking. Tracking plays a vital role in locating vehicles during theft scenarios, making it an important feature.

4. Lack of Engine Immobilization Feature

The system lacks a mechanism to physically prevent unauthorized vehicle operation. Engine immobilization is a critical component in modern anti-theft systems.

5. Limited Scope to Cyber-Level Security

The system mainly focuses on cyber-level threats and does not address physical security aspects such as intrusion detection using sensors.

6. Lack of Integrated Security Framework

The base paper presents a standalone solution without integrating detection, alerting, tracking, and prevention. This highlights the need for a comprehensive vehicle security framework.

2.4 Summary

The base paper introduces an intrusion detection system that utilizes CAN frame-aware features such as message identifiers, timing intervals, and payload patterns to identify abnormal activities within the network. By employing machine learning and deep learning techniques, the system is capable of detecting various cyber-attacks, including spoofing, replay, and Denial-of-Service (DoS) attacks.

The review highlights the strengths of the approach, particularly its ability to analyze both structural and temporal characteristics of CAN messages. However, it also identifies limitations such as the lack of preventive mechanisms, absence of real-time alert systems, and no integration with tracking or engine immobilization features.

Overall, the literature survey shows that existing solutions focus mainly on detection and do not provide a complete vehicle security framework. These observations emphasize the need for an integrated system combining detection, alerting, tracking, and control.

CHAPTER 3

System Design and Architecture

3.1 Introduction

The system design of the proposed vehicle anti-theft system focuses on developing an integrated framework to enhance vehicle security through effective monitoring and control. The design combines sensing, processing, communication, and control units to ensure reliable and coordinated operation of the system.

The system is implemented using ESP32 microcontrollers as the main processing units, along with CAN bus communication to enable efficient data exchange between different nodes. Additional components such as sensors, communication modules, and control units are incorporated to detect unauthorized access and perform necessary actions.

The architecture is designed to support real-time monitoring and user interaction through a web-based interface, allowing control of vehicle functions such as locking, unlocking, and parking. This ensures that the system not only detects potential threats but also enables immediate response.

Overall, the design aims to provide a secure, reliable, and efficient solution by integrating multiple technologies within a single system architecture, thereby improving vehicle protection against modern security threats.

3.2 Block Diagram

The block diagram of the proposed vehicle anti-theft system illustrates the overall architecture and interaction between different hardware modules used in the system. It provides a clear representation of how sensing, communication, processing, and control units are interconnected to achieve vehicle security.

The system is divided into two main sections: the vehicle module and the owner's handheld module. The vehicle module consists of multiple ESP32

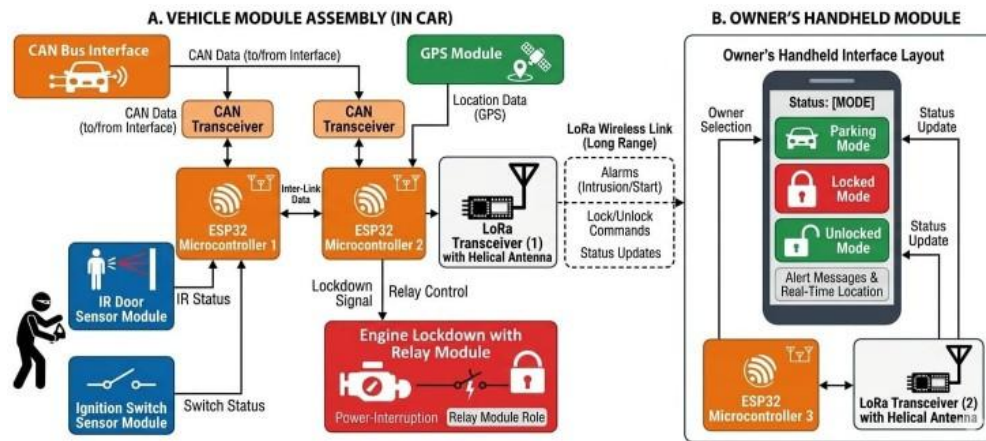


Figure 3.1: Block Diagram of the Proposed Vehicle Anti-Theft System microcontrollers, CAN transceivers, sensors, and control components, while the owner's module enables user interaction and monitoring.

In the vehicle module, ESP32 Microcontroller 1 is responsible for handling sensor inputs. It receives data from the IR door sensor module, which detects intrusion, and the ignition switch sensor module, which monitors the status of the vehicle ignition. This data is processed and transmitted to ESP32 Microcontroller 2.

ESP32 Microcontroller 2 acts as an intermediate processing and communication unit. It is connected to CAN transceivers that enable communication with the CAN bus interface. It also interfaces with the GPS module to obtain real-time location data of the vehicle. Additionally, it controls the relay module used for engine lockdown by generating appropriate control signals.

The CAN transceivers (MCP2551) are used to facilitate communication between ESP32 nodes through the CAN bus. This ensures reliable and efficient data transfer within the system, similar to communication in real automotive networks.

For long-range communication, a LoRa transceiver is connected to ESP32 Microcontroller 2. This module transmits alerts, system status, and control data to the owner's handheld module. The use of LoRa enables communication over longer distances with low power consumption.

On the user side, ESP32 Microcontroller 3 is connected to another LoRa transceiver, forming a wireless communication link with the vehicle module. This allows the user to receive alerts such as intrusion detection and engine

status, as well as send commands such as lock, unlock, and parking mode.

The engine lockdown mechanism is implemented using a relay module. When an intrusion is detected or a lock command is issued, the relay interrupts the vehicle's ignition system, preventing unauthorized engine operation.

Overall, the block diagram clearly represents the integration of sensing, processing, communication, and control modules. It demonstrates how the system detects unauthorized access, communicates information across modules, and enables user-controlled actions to enhance vehicle security.

3.3 Circuit Diagram

The circuit diagram of the proposed vehicle anti-theft system provides a detailed representation of the hardware connections between various modules. It illustrates how sensing, communication, and control components are integrated to achieve effective vehicle security.

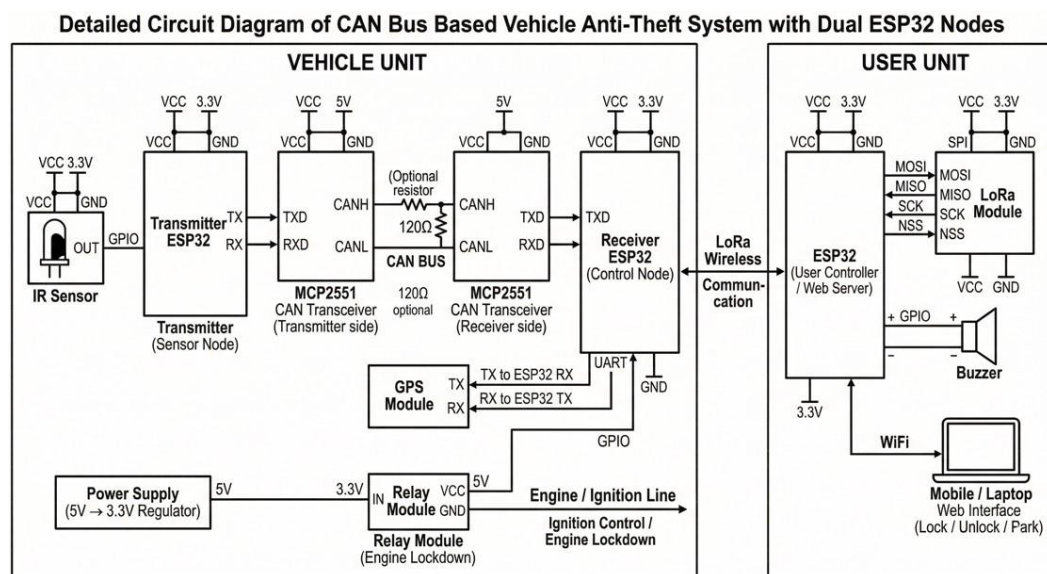


Figure 3.2: Detailed Circuit Diagram of CAN Bus Based Vehicle Anti-Theft System

The system is divided into two main sections, namely the Vehicle Unit and the User Unit. The Vehicle Unit consists of two ESP32 microcontrollers, CAN transceivers, sensors, and control modules, while the User Unit enables monitoring and control through wireless communication.

In the Vehicle Unit, the Transmitter ESP32 is connected to the IR sensor,

which detects intrusion or unauthorized access. The sensor output is given to the GPIO pin of the ESP32. The transmitter ESP32 communicates with the CAN transceiver (MCP2551) using TX and RX lines. The CAN transceiver converts the data into CAN protocol format and transmits it over the CAN bus through CANH and CANL lines.

The CAN bus connects the transmitter and receiver nodes, allowing reliable data communication between the two ESP32 modules. A termination resistor of 120 ohms is used across CANH and CANL to ensure proper signal integrity and reduce communication errors.

The Receiver ESP32 receives CAN messages through another MCP2551 transceiver. It processes the received data and performs control actions. The receiver ESP32 is connected to a GPS module using UART communication (TX and RX), which provides real-time location data of the vehicle.

The receiver ESP32 also controls a relay module used for engine lockdown.

The relay is connected to a GPIO pin of the ESP32 and interrupts the ignition line of the vehicle when activated, preventing unauthorized engine operation.

A regulated power supply is used to provide both 5V and 3.3V to different components. The ESP32 operates at 3.3V, while modules such as MCP2551, GPS, and relay use 5V supply.

In the User Unit, an ESP32 microcontroller acts as a web server and user interface controller. It is connected to a LoRa module using SPI communication (MOSI, MISO, SCK, NSS). The LoRa module enables long-range wireless communication between the vehicle unit and the user unit.

A buzzer is connected to the ESP32 to provide audible alerts in case of intrusion or system warnings. The ESP32 also connects to a mobile or laptop via WiFi, allowing the user to access a web interface for monitoring vehicle status and controlling functions such as lock, unlock, and parking mode.

The communication between the Vehicle Unit and User Unit is achieved through LoRa wireless technology, enabling long-range and low-power data transmission. Overall, the circuit diagram demonstrates the complete integration of sensing, processing, communication, and control modules in the proposed system.

3.4 Hardware Components

The proposed vehicle anti-theft system is developed using a combination of hardware components that perform sensing, communication, processing, and control functions. Each component plays a specific role in ensuring the proper operation of the system. The details of the hardware components used in the system are described in the following subsections.

3.4.1 ESP32

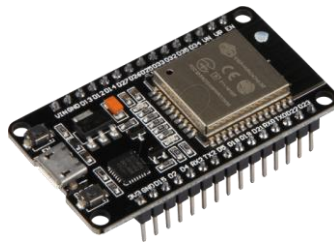


Figure 3.3: ESP32 Microcontroller

The ESP32 microcontroller is the core component of the system and is responsible for processing, communication, and control operations. In this project, multiple ESP32 boards are used as transmitter, receiver, and main controller nodes. The ESP32 supports WiFi communication, which enables the implementation of a web-based interface for user interaction. It also provides multiple GPIO pins for interfacing with sensors and other modules.

3.4.2 MCP2551 CAN Transceiver

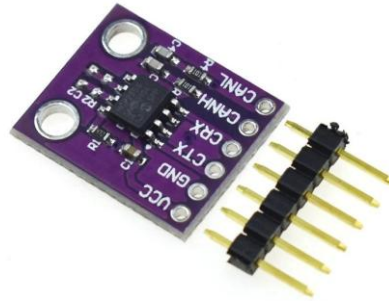


Figure 3.4: MCP2551 CAN Transceiver Module

The MCP2551 CAN transceiver is used to enable CAN bus communication between ESP32 nodes. It converts digital signals from the microcontroller into CAN bus signals and vice versa. Two MCP2551 modules are used in the system to establish communication between transmitter and receiver nodes through CANH and CANL lines, ensuring reliable data transfer.

3.4.3 LoRa Module



Figure 3.5: LoRa Communication Module

The LoRa module is used for long-range wireless communication between system nodes. It enables data transmission over larger distances with low power consumption. The module is interfaced with the ESP32 using SPI communication.

3.4.4 GPS Module



Figure 3.6: GPS Module (NEO-6M)

The GPS module is used to obtain real-time location information of the vehicle. It communicates with the ESP32 through UART and provides latitude and longitude data, which can be used for monitoring and tracking purposes.

3.4.5 Relay Module

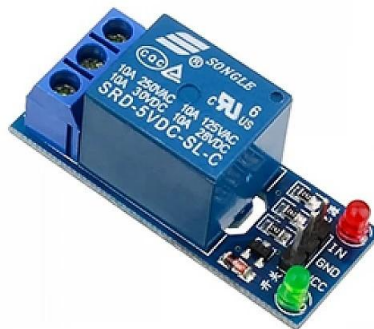


Figure 3.7: Relay Module

The relay module is used to control the vehicle ignition system. It acts as a switch that can enable or disable the engine based on the control signals received from the ESP32. This feature allows the system to prevent unauthorized vehicle operation.

3.4.6 IR Sensor

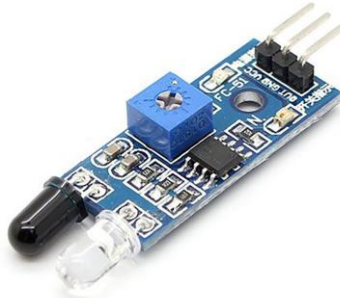


Figure 3.8: Infrared (IR) Sensor Module

The IR sensor is used for intrusion detection in the system. It detects the presence of an object or movement near the vehicle and generates a signal when unauthorized access is identified. This signal is sent to the ESP32 for further processing and action.

3.4.7 Buzzer



Figure 3.9: Buzzer Module

The buzzer is used to provide an audible alert during intrusion or abnormal conditions. It is activated by the ESP32 to notify nearby users about unauthorized access.

3.5 Software

The development of the proposed vehicle anti-theft system involves the use of various software tools for programming, simulation, and system control.

The Arduino IDE is used for writing, compiling, and uploading the program code to the ESP32 microcontrollers. It provides a simple and efficient environment for developing embedded applications and interfacing with different hardware modules.

Embedded C is used as the programming language to implement the system logic, including sensor data processing, communication between nodes, and control operations. The code is developed to handle real-time events such as intrusion detection and user commands.

A web-based interface is implemented using the ESP32's built-in WiFi capability. The ESP32 acts as a web server and generates a local IP address, allowing the user to access the system through a browser. Through this interface, the user can monitor system status and perform operations such as locking, unlocking, and parking.

These software components work together to ensure proper functioning, control, and user interaction within the system.

3.6 Summary

This chapter presented the system design of the proposed vehicle anti-theft system, explaining the overall architecture and working of the system. The block diagram showed the interaction between different modules, while the circuit diagram illustrated the hardware connections and communication between ESP32 nodes using CAN bus.

The chapter also described the hardware and software components used in the system. Key modules such as ESP32, sensors, CAN transceivers, and control units were explained, along with the use of Arduino IDE and a web-based interface for system control. Overall, the chapter provided a clear view of the system implementation.

CHAPTER 4

Implementation and Methodology

4.1 Introduction

This chapter presents the implementation and operational aspects of the proposed vehicle anti-theft system. It explains how the system is developed by integrating various hardware and software components to provide an effective vehicle security solution. The focus of this chapter is to describe how the designed system is practically realized.

Modern vehicles rely heavily on electronic systems and communication networks for efficient operation. While these advancements have improved performance and safety, they have also introduced security challenges such as unauthorized access and cyber-attacks. Therefore, it is important to develop systems that can detect and respond to such threats effectively.

The proposed system addresses these challenges by combining CAN bus communication, sensor-based intrusion detection, and control mechanisms. ESP32 microcontrollers are used for processing and communication between different modules, while additional components enable detection and response to unauthorized activities.

A web-based interface is implemented to allow the user to monitor and control the system in real time. The ESP32 acts as a web server, providing access through a local IP address. This enables the user to view system status and perform actions such as locking, unlocking, and parking.

This chapter further describes the methodology, flow of operations, and working principle of the system, providing a clear understanding of how the different components interact to ensure effective vehicle security.

4.2 Methodology

The methodology of the proposed vehicle anti-theft system describes the complete operational flow of the system, starting from monitoring and intrusion detection to communication, user interaction, and control actions. The system is designed to operate continuously in real time to ensure effective vehicle security and quick response to unauthorized activities.

Initially, the system operates in a monitoring state, where the IR sensor continuously observes the surroundings of the vehicle. The sensor detects any movement or intrusion near the vehicle and generates a signal when such activity is identified. This signal is transmitted to the transmitter ESP32, which serves as the first stage of processing in the system. Continuous monitoring ensures that the system remains active at all times and can respond immediately to any suspicious activity.

Once an intrusion is detected, the transmitter ESP32 processes the input signal received from the sensor. The data is formatted into a suitable structure for communication over the CAN bus. The use of CAN communication provides a reliable and efficient method for transmitting data between different nodes in the system, ensuring minimal delay and high accuracy.

The processed data is then transmitted through the CAN bus using the MCP2551 CAN transceiver. The transceiver converts the data into CAN protocol format and ensures proper communication over the bus. This step is critical for maintaining synchronization between different ESP32 nodes and enabling robust data exchange.

At the receiver side, another ESP32 connected to a CAN transceiver receives the transmitted data. The receiver ESP32 decodes the incoming CAN message and extracts the relevant information. This data is then forwarded to the main controller ESP32 for further processing. The separation of transmitter and receiver nodes improves the modularity and scalability of the system.

The main controller ESP32 acts as the central processing unit of the system. It is responsible for managing communication, processing received data, and controlling system operations. One of its key features is hosting a

web server using its built-in WiFi capability. The ESP32 generates a local IP address, allowing the user to access the system through a web browser.

The web-based interface plays an important role in user interaction. It displays real-time system status, including intrusion alerts and system conditions. The interface provides control options such as locking, unlocking, and parking mode selection. This allows the user to monitor and control the vehicle remotely in a simple and efficient manner.

The overall sequence of operations involved in the system is summarized in Table 4.1.

Table 4.1: System Operation Steps

S.No	Step	Description
1	Monitoring	IR sensor continuously monitors surroundings
2	Detection	Intrusion is detected and signal is generated
3	Processing	ESP32 processes sensor data
4	Communication	Data transmitted via CAN bus
5	Reception	Receiver ESP32 receives data
6	Display	Status shown on web interface
7	Control	User selects lock/unlock/park
8	Action	Relay and buzzer perform actions

After the data is processed by the main controller, appropriate actions are performed based on the system condition or user input. If an intrusion is detected, the system activates the buzzer to provide an audible alert. At the same time, the system status is updated on the web interface, allowing the user to take necessary action.

The user can interact with the system through the web interface by selecting options such as lock, unlock, or parking mode. These commands are sent to the ESP32, which processes them and controls the relay module accordingly. The relay module is used to control the vehicle ignition system, enabling or disabling the engine based on the selected command.

In addition to intrusion detection and control, the system also incorporates a GPS module to provide location information. The GPS module continuously tracks the position of the vehicle and provides real-time location data. This feature enhances the security of the system by allowing the user to locate the

vehicle in case of theft or unauthorized movement.

The system also supports long-range communication through the LoRa module, which enables communication between nodes over larger distances. This improves the flexibility and scalability of the system, especially in real-world applications where long-range communication may be required.

The roles of different modules involved in the system are summarized in Table 4.2.

Table 4.2: Functional Roles of System Modules

Module	Function
IR Sensor	Detects intrusion near vehicle
ESP32 (Transmitter)	Processes sensor data and sends via CAN
ESP32 (Receiver)	Receives CAN data
ESP32 (Controller)	Hosts web server and controls system
MCP2551	Enables CAN communication
GPS Module	Provides location information
LoRa Module	Enables long-range communication
Relay Module	Controls engine ignition
Buzzer	Provides alert during intrusion

The system operates in a continuous loop, ensuring that monitoring, communication, and control processes are carried out without interruption. This continuous operation enhances the reliability and effectiveness of the system in providing vehicle security.

Overall, the methodology demonstrates how different components are integrated to achieve a complete vehicle anti-theft system. By combining intrusion detection, communication, user interaction, and control mechanisms, the system provides a comprehensive solution for improving vehicle safety and preventing unauthorized access.

4.3 Flowchart

The flowchart of the proposed vehicle anti-theft system represents the sequence of operations performed by the system, starting from initialization to monitoring, intrusion detection, communication, and control actions. It provides a clear understanding of how different modules interact to ensure effective vehicle security.

Initially, the system begins with the initialization phase, where all hardware components and communication modules are configured. This includes initializing the ESP32 microcontrollers, CAN communication using MCP2551, and connected sensors such as the IR sensor. Once initialization is complete, the system enters the monitoring state.

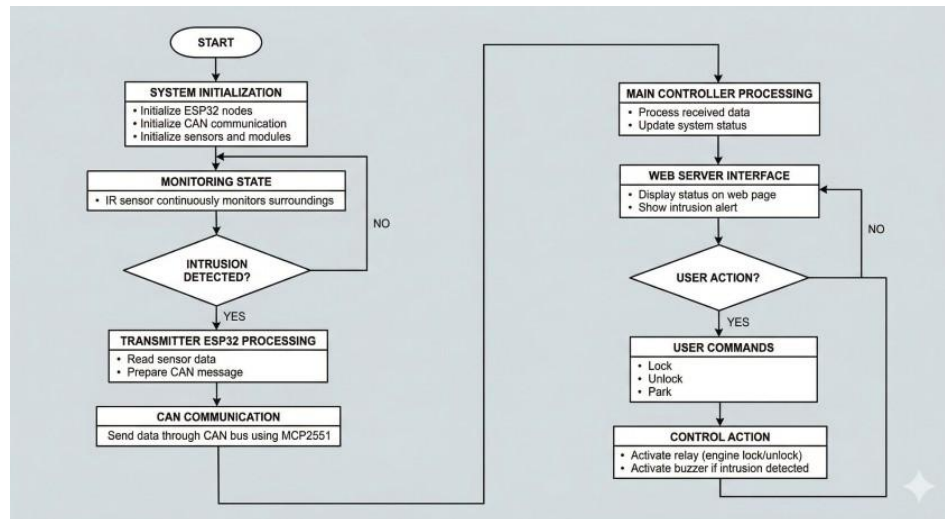


Figure 4.1: Flowchart of the Proposed Vehicle Anti-Theft System

In the monitoring state, the IR sensor continuously observes the surroundings of the vehicle to detect any unauthorized access or movement. If no intrusion is detected, the system remains in this state, ensuring continuous surveillance. This loop allows the system to operate continuously without interruption.

When an intrusion is detected, the system moves to the processing stage. The transmitter ESP32 reads the sensor data and prepares a CAN message. This message is then transmitted through the CAN bus using the MCP2551 transceiver, ensuring reliable and fast communication between system nodes.

The transmitted data is received by the receiver ESP32, which forwards the information to the main controller ESP32. The main controller processes the received data and updates the system status accordingly. This status is then displayed on the web server interface, allowing the user to monitor the system in real time.

The web interface provides information such as intrusion alerts and system conditions. At this stage, the system checks for user interaction. If no user

action is detected, the system continues to operate in monitoring mode.

If the user provides input through the web interface, such as selecting lock, unlock, or parking mode, the system processes these commands and performs appropriate control actions. The relay module is activated to control the vehicle ignition system, enabling or disabling the engine based on the selected command.

Additionally, the buzzer is activated when an intrusion is detected to provide an audible alert. After executing the required action, the system returns to the monitoring state, ensuring continuous operation.

Thus, the flowchart clearly illustrates the complete operational flow of the system, including detection, communication, user interaction, and control. It helps in understanding how the system responds to different conditions and maintains vehicle security effectively.

4.4 System Operation

The system operation describes the working of the proposed vehicle anti-theft system in real-time conditions. It explains how different components interact with each other to detect intrusion, communicate information, and perform control actions based on system conditions and user input.

Initially, when the system is powered on, all modules including ESP32 nodes, CAN transceivers, sensors, and communication interfaces are initialized. Once initialization is complete, the system enters the monitoring mode, where it continuously observes the vehicle environment for any suspicious activity.

In the monitoring phase, the IR sensor plays a key role in detecting unauthorized access. It continuously scans the surroundings of the vehicle. If no intrusion is detected, the system remains in this state, ensuring continuous surveillance. This loop ensures that the system is always active and ready to respond.

When an intrusion is detected, the IR sensor sends a signal to the transmitter ESP32. The transmitter ESP32 reads the sensor data and processes it to generate a CAN message. This message is then transmitted over the CAN bus using the MCP2551 transceiver.

At the receiver side, the ESP32 connected to another CAN transceiver receives the transmitted data. The receiver ESP32 decodes the message and forwards it to the main controller ESP32. The main controller processes this information and updates the system status accordingly.

The main controller ESP32 hosts a web server that allows the user to interact with the system through a web interface. It generates a local IP address, which can be accessed using a mobile phone or computer. The web interface displays system status, including intrusion alerts and system conditions.

When an intrusion is detected, the system immediately updates the web interface and activates the buzzer to provide an audible alert. This ensures that both visual and audible notifications are provided to the user.

The system also allows user interaction through the web interface. The user can select different modes such as Lock, Unlock, and Parking. These commands are sent to the ESP32 and processed accordingly.

The different operating modes of the system and their functions are summarized in Table 4.3.

Table 4.3: System Modes and Corresponding Actions

S.No	Mode	Description
1	Parking Mode	System continuously monitors for intrusion and generates alerts when unauthorized access is detected
2	Lock Mode	Engine is disabled using relay to prevent unauthorized vehicle operation
3	Unlock Mode	Vehicle operates normally and system allows regular usage

In Lock mode, the system restricts vehicle operation by disabling the engine through the relay module. In Unlock mode, the system allows normal vehicle operation. In Parking mode, the system remains highly sensitive to intrusion and continuously monitors for unauthorized activity.

Based on the selected command, the relay module is activated or deactivated to control the vehicle ignition system. This provides a physical layer of security by preventing unauthorized engine operation.

In addition, the GPS module continuously tracks the location of the vehicle

and provides real-time position information. This feature enhances the system by allowing the user to locate the vehicle in case of theft.

The system operates in a continuous loop, returning to the monitoring state after each operation. This ensures that detection, communication, and control processes are performed continuously without interruption.

Overall, the system operation demonstrates how multiple modules work together to provide a comprehensive vehicle security solution. By combining intrusion detection, communication through CAN bus, user interaction via web interface, and control through relay mechanisms, the system effectively enhances vehicle safety and prevents unauthorized access.

4.5 Summary

This chapter presented the implementation and operational methodology of the proposed vehicle anti-theft system. It explained how the system is developed by integrating multiple hardware components such as ESP32 microcontrollers, CAN transceivers, sensors, and control modules to achieve effective vehicle security.

The methodology section described the step-by-step operation of the system, including monitoring, intrusion detection, data communication through CAN bus, and control actions. The flowchart provided a clear representation of the system workflow, illustrating how different stages of operation are carried out in a structured manner.

The system operation section explained the real-time functioning of the system, including user interaction through the web interface and execution of control actions such as locking, unlocking, and intrusion alerting. The inclusion of system modes and their corresponding actions further highlighted the practical working of the system.

Overall, this chapter demonstrated how the proposed system integrates detection, communication, user interaction, and control mechanisms to provide a reliable and efficient vehicle anti-theft solution.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents a detailed discussion of the results obtained from the implementation of the proposed CAN Bus based vehicle anti-theft system with engine lockdown and GPS tracking. The purpose of this chapter is to evaluate the performance of the system under different operating conditions and to verify whether the objectives of the project have been successfully achieved.

The developed system integrates multiple technologies such as CAN communication, ESP32 microcontrollers, LoRa wireless communication, GPS tracking, and sensor-based intrusion detection. The results are analyzed by testing the system in various modes of operation, namely Lock Mode, Unlock Mode, and Parking Mode. Each mode represents a specific security condition of the vehicle and demonstrates how the system responds to user commands and external events.

The chapter also highlights how effectively the system performs real-time monitoring, communication, and control actions. The experimental results are supported by images captured during testing, which illustrate the practical implementation and working of the system. Further, the results are analyzed to understand the behavior of individual modules and their interaction within the system.

In addition to performance evaluation, this chapter also includes a detailed analysis of system behavior and an evaluation based on various quality factors such as safety, cost, sustainability, and reliability. These discussions help in assessing the feasibility and real-world applicability of the proposed solution. Overall, this chapter provides a comprehensive understanding of the system's performance and validates the effectiveness of the proposed vehicle anti-theft

system.

5.2 Overview of Results

The developed system was tested in real-time conditions to evaluate its performance under different operating modes. The results demonstrate that the system successfully performs intrusion detection, communication through CAN bus and LoRa, and control actions such as engine lockdown and alert generation. The working of the system is analyzed based on three primary modes of operation.

Lock Mode

In Lock Mode, the system provides maximum security by preventing unauthorized operation of the vehicle. When this mode is activated through the web interface, the user command is transmitted to the vehicle unit via LoRa communication. The receiver ESP32 processes this command and activates the relay module connected to the ignition system.

The relay module interrupts the ignition line, effectively disabling the engine. This ensures that even if an intruder attempts to start the vehicle, the engine will not respond. During testing, it was observed that the system successfully blocked engine operation when Lock Mode was enabled.

Additionally, the system remains alert for any intrusion attempts. If the IR sensor detects unauthorized access while in Lock Mode, the system generates an alert and notifies the user through the web interface. This dual-layer protection mechanism enhances the overall security of the vehicle.

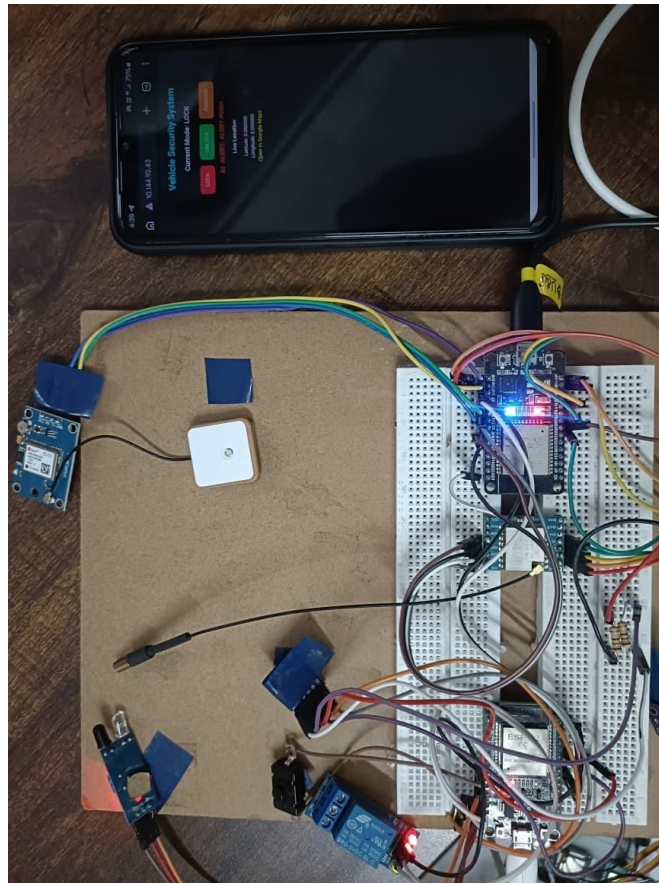


Figure 5.1: System Operation in Lock Mode

Unlock Mode

In Unlock Mode, the system allows normal vehicle operation by deactivating the engine lockdown mechanism. When the user selects Unlock Mode through the web interface, the command is transmitted via LoRa to the vehicle unit. The receiver ESP32 processes this command and deactivates the relay module, restoring the ignition connection.

This mode is used when the authorized user wants to operate the vehicle normally. During testing, it was observed that the vehicle operated without any restrictions once Unlock Mode was activated. The system ensured smooth transition between Lock and Unlock states without delay.

Although the system allows normal operation in this mode, it still maintains basic monitoring capabilities. The communication between modules remains active, and the system is ready to respond to any new commands or changes in state. This ensures continuous system readiness and reliability.

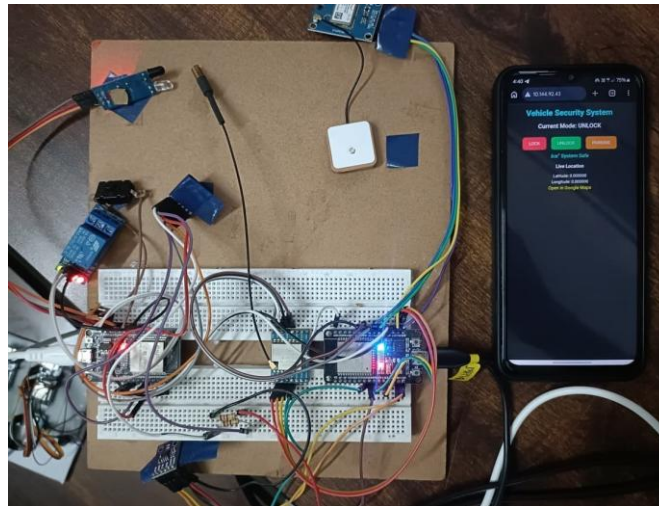


Figure 5.2: System Operation in Unlock Mode

Parking Mode

Parking Mode is designed for monitoring the vehicle when it is stationary. In this mode, the system continuously observes the surroundings using the IR sensor to detect any unauthorized movement or intrusion.

When an intrusion is detected, the IR sensor sends a signal to the transmitter ESP32. This data is transmitted via CAN bus to the receiver ESP32, which then processes the information and triggers an alert. The system activates the buzzer to provide an immediate warning and simultaneously sends a notification to the user through the web interface.

During testing, it was observed that the system responded quickly to intrusion events and successfully alerted the user. The integration of CAN communication ensures reliable data transfer, while LoRa communication enables long-range alert transmission.

This mode is particularly useful for enhancing vehicle security in parking areas, where the vehicle is left unattended. It provides continuous surveillance and immediate response, reducing the risk of theft or damage.

Overall, the results demonstrate that the system operates effectively across all modes and provides a comprehensive vehicle security solution.

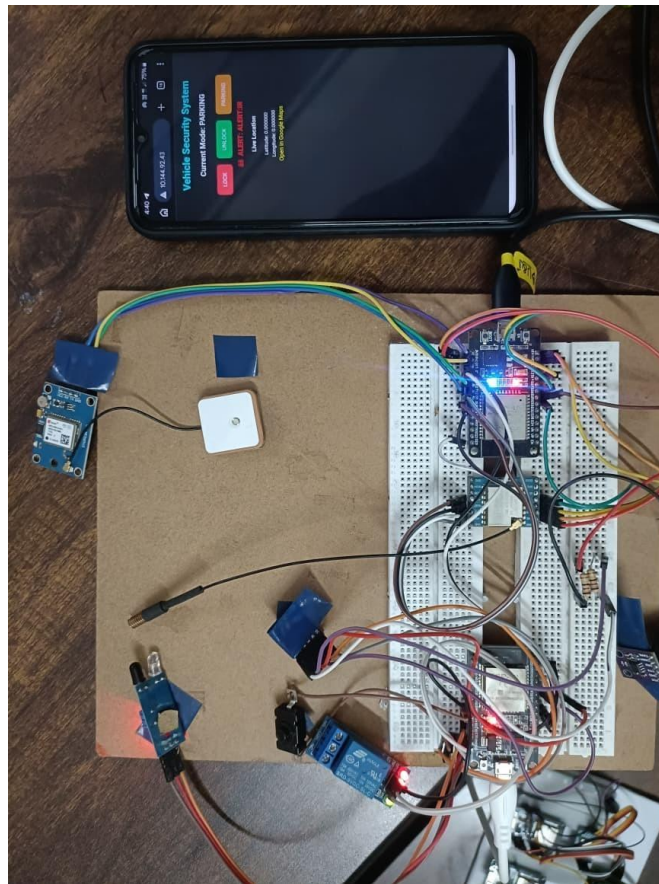


Figure 5.3: System Operation in Parking Mode

5.3 Analysis and Interpretation

The results obtained from the experimental setup confirm that the proposed system performs reliably in real-time conditions. The IR sensor effectively detects intrusion and provides accurate input to the system. The use of CAN bus ensures fast and reliable communication between the transmitter and receiver ESP32 nodes.

The integration of LoRa communication allows long-range data transmission, enabling the user to receive alerts and send commands from a remote location. The relay module successfully performs engine lockdown, demonstrating the practical applicability of the system.

The coordination between sensing, communication, and control modules shows that the system is well-designed and capable of handling real-world scenarios. The GPS module further enhances the system by providing location tracking, making it more effective in case of theft.

To better understand the effectiveness of the proposed system, a comparison with traditional vehicle security systems is presented in Table 5.1.

Table 5.1: Comparison Between Traditional and Proposed System

S.No	Traditional System	Proposed System
1	Basic alarm system only	Intrusion detection with alert and control
2	No remote monitoring	Web-based remote monitoring via WiFi
3	No real-time tracking	GPS-based real-time location tracking
4	No engine immobilization	Engine lockdown using relay module
5	Limited communication	CAN + LoRa communication for reliability
6	Works only locally	Supports long-range communication
7	Low security level	Multi-layer security system

From the comparison, it is evident that the proposed system provides a more advanced and comprehensive solution compared to traditional systems.

It not only detects intrusion but also enables tracking, remote monitoring, and preventive control actions.

Overall, the analysis indicates that the system achieves its intended functionality and provides a reliable and efficient vehicle security solution.

5.4 Evaluation of Quality Factors

In this section, the results are evaluated based on several quality factors:

- **Environment:** The system contributes to reducing vehicle theft, indirectly preventing resource loss and minimizing environmental impact associated with vehicle replacement and damage.
- **Sustainability:** The system is designed using low-power and cost-effective components such as ESP32 and LoRa, making it suitable for long-term and scalable deployment.
- **Safety:** The system enhances safety by preventing unauthorized vehicle usage through engine lockdown and providing real-time alerts to the user.
- **Ethics:** The system ensures ethical use by considering user privacy and controlled access to vehicle data and tracking information.
- **Cost:** The system is cost-effective compared to traditional vehicle security systems, using affordable hardware components and simple implementation techniques.
- **Type:** The project produces a practical embedded system solution aligned with automotive security applications and real-world use.
- **Standards:** The system follows basic automotive communication principles using CAN bus (ISO 11898) and considers safety practices in embedded system design.

The evaluation shows that the system meets essential quality requirements and provides a reliable, efficient, and practical solution for vehicle security.

5.5 Summary

This chapter presented a detailed evaluation of the proposed vehicle anti-theft system through practical testing and analysis. The system was tested under different operating modes, and the results confirmed its ability to detect intrusion, communicate effectively, and control vehicle operation.

The analysis demonstrated that the integration of CAN bus, LoRa communication, and embedded control mechanisms provides a reliable and efficient security solution. The evaluation based on quality factors further validated the system's effectiveness, safety, and cost-efficiency.

Overall, the results confirm that the proposed system successfully meets its objectives and provides a strong foundation for future enhancements in vehicle security systems.

CHAPTER 6

Conclusions and Future Scope

6.1 Conclusions

The project titled “CAN Bus Based Vehicle Anti-Theft System with Engine Lockdown and GPS” was successfully developed as a smart and practical prototype to enhance vehicle security. The system effectively integrates intrusion detection, real-time alerting, engine immobilization, and GPS-based tracking using ESP32 microcontrollers, CAN transceivers, LoRa communication, and sensor modules. The multi-layered security approach ensures early detection of unauthorized access and allows the user to take timely action through a web-based interface. Compared to conventional systems, the proposed solution provides a more advanced and reliable mechanism by combining both preventive and reactive security features.

The project also demonstrates strong relevance to modern automotive systems by incorporating CAN bus communication and modular embedded design. While the prototype proves to be cost-effective and efficient, certain limitations such as safety considerations in engine control and secure communication need to be addressed for real-world deployment. Overall, the system successfully achieves its objectives and serves as a strong foundation for future development in intelligent vehicle security systems, contributing to safer and more secure transportation.

6.2 Future Scope of Work

The proposed vehicle anti-theft system offers significant scope for future enhancement to make it more robust, intelligent, and suitable for real-world deployment. One major improvement would be replacing the relay-based engine lockdown with a safer ECU-level authorization mechanism, ensuring secure and controlled engine start. Future systems can also integrate real-time

CAN message analysis to detect abnormal vehicle behavior such as spoofed commands or unauthorized ECU communication. Additionally, incorporating cloud connectivity and mobile application support would enable real-time monitoring, data logging, geofencing, and remote control through user-friendly interfaces, enhancing usability and system effectiveness.

Further advancements can be made by improving intrusion detection using multiple sensors such as vibration, tilt, and ultrasonic sensors to increase accuracy and reduce false alerts. The system can also be enhanced with multi-factor authentication methods such as RFID, biometrics, or smartphone-based access to strengthen security. Implementing cybersecurity measures like encryption, secure communication, and authentication protocols will make the system more reliable against electronic attacks. Overall, with the integration of intelligent algorithms, advanced hardware, and secure communication, the project can evolve into a comprehensive next-generation vehicle security solution suitable for modern connected vehicles.

CHAPTER 7

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

1. Program Outcomes (POs) and Program Specific Outcomes (PSOs)

The project contributes to the attainment of the following Program Outcomes:

PO1: Engineering Knowledge

Applied knowledge of embedded systems, CAN bus communication, and wireless technologies such as LoRa and GPS to design and develop the vehicle anti-theft system.

PO2: Problem Analysis

Analyzed security vulnerabilities in vehicle communication systems, particularly in CAN networks, and identified issues such as intrusion, unauthorized access, and lack of real-time response mechanisms.

PO3: Design/Development of Solutions

Designed and developed a complete vehicle anti-theft system integrating intrusion detection, CAN communication, web-based control, and engine lockdown mechanisms while considering safety and practical implementation.

PO4: Conduct Investigations of Complex Problems

Investigated system behavior by testing intrusion scenarios, analyzing sensor responses, and validating communication between multiple ESP32 nodes and CAN modules.

PO5: Engineering Tool Usage

Utilized modern tools such as Arduino IDE, embedded C programming, ESP32 platform, and hardware modules like MCP2551, GPS, and LoRa for system development and testing.

PO6: The Engineer and The World

Considered the impact of vehicle security systems on safety, prevention of theft, and protection of user property, contributing to societal benefits and secure transportation.

PO7: Ethics

Ensured ethical use of tracking and control features by considering user privacy, data security, and responsible system usage.

PO8: Individual and Team Work Worked collaboratively in a team environment, dividing tasks such as hardware integration, software development, and documentation effectively.

PO9: Communication

Prepared technical reports, documentation, and presentations to clearly explain the system design, methodology, and results.

PO10: Project Management and Finance

Managed project planning, component selection, and implementation within available resources, ensuring cost-effective system development.

PO11: Life-Long Learning

Explored emerging technologies such as IoT, CAN security, and wireless communication, demonstrating the ability to learn and adapt to new concepts.

2. Program Specific Outcomes (PSOs)**PSO 01: Application of Core Electronics and Communication Concepts.**

Applied domain-specific knowledge in embedded systems and communication protocols to design and implement a CAN Bus-based vehicle anti-theft system using ESP32 microcontrollers, sensors, and communication modules.

3. Sustainable Development Goals (SDGs) The project supports the following Sustainable Development Goals:

SDG 09: Industry, Innovation and Infrastructure

The proposed system contributes to innovation in automotive security by

integrating modern communication technologies such as CAN bus, IoT, and wireless modules to enhance vehicle safety and reliability.

SDG 11: Sustainable Cities and Communities

The system improves urban safety by reducing vehicle theft and enabling secure transportation systems, thereby contributing to safer and more sustainable cities.

Overall Impact

The CAN Bus Based Vehicle Anti-Theft System with Engine Lockdown and GPS effectively maps to multiple Program Outcomes (PO1–PO11) by demonstrating engineering knowledge, problem analysis, system design, modern tool usage, teamwork, ethical responsibility, communication skills, and project management capabilities. The project also aligns with the Program Specific Outcomes (PSOs) by applying core concepts of embedded systems and communication engineering to design and develop a real-time vehicle security system using ESP32 microcontrollers, CAN bus communication, LoRa wireless technology, GPS tracking, sensor interfacing, and web-based monitoring.

In addition, the project supports important Sustainable Development Goals (SDGs), particularly SDG 9 and SDG 11, as it promotes innovation in automotive security systems and contributes to safer and more secure communities by reducing vehicle theft. Thus, the project not only fulfills academic learning outcomes and program-specific objectives but also addresses practical societal needs and enhances the reliability and safety of modern transportation systems.

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